AQUEN FLUID TECHNOLOGIES LLP

Complete Water & Wastewater Treatment Solutions

Design | Manufacturing | Supply | Installation | Commissioning | AMC

FRP PACKAGE SEWAGE TREATMENT PLANT (STP)

TECHNICAL CATALOGUE

2 KLD TO 60 KLD | OPERATION HOURS : 22 HRS

- Compact & Modular Design
- No Civil Tanks (Complete Package)
- Quick Installation & Relocation
- Low Operation & Maintenance
- Odour Free Operation
- Reliable & Stable Performance
- Suitable for Residential, Commercial, Institutional & Industrial Applications
- Treated Water Tank for Flushing Purpose (TTP)

CLEANER WATER GREENER TOMORROW



1. FRP STP Installation – 60 KLD (Tank Placement)



2. FRP STP Installation – 50 KLD (Tank Placement with Hydra)



3. Gravel Bed Preparation (For FRP Tanks)



4. Piping & Connection (Installation in Progress)



5. FRP STP Installation – 100 KLD (Example Larger Capacity)



6. PCC Base Preparation (Before Tank Placement)

PROCESS FLOW DIAGRAM – FRP PACKAGE STP (NO CIVIL TANKS)



TYPICAL OUTLET WATER QUALITY (For Flushing Reuse)

Parameter	Typical
pH	6.5 – 8.0
BOD (mg/L)	< 10
COD (mg/L)	< 30
TSS (mg/L)	< 10
Oil & Grease (mg/L)	< 5
Ammonia – N (mg/L)	< 5
Total Nitrogen (mg/L)	< 10
Fecal Coliform (MPN/100 mL)	< 10^2

TYPICAL INLET SEWAGE PARAMETERS (Domestic Sewage)

Parameter	Typical
pH	6.5 – 8.5
BOD (mg/L)	200 – 300
COD (mg/L)	400 – 600
TSS (mg/L)	200 – 300
Oil & Grease (mg/L)	10 – 20
Ammonia – N (mg/L)	20 – 40
Total Nitrogen (mg/L)	30 – 50
Fecal Coliform (MPN/100 mL)	10^5 – 10^6

TECHNICAL SPECIFICATIONS – FRP PACKAGE STP (2 KLD TO 60 KLD, OPERATION HOURS : 22 HRS)

Parameter	Unit	2 KLD	5 KLD	10 KLD	15 KLD	20 KLD	25 KLD	30 KLD	50 KLD	60 KLD
Design Flow	KLD	2	5	10	15	20	25	30	50	60
Operation Hours	Hrs	22	22	22	22	22	22	22	22	22
Average Flow	m³/hr	0.09	0.23	0.45	0.68	0.91	1.14	2.27	2.23	2.73
No. of FRP Tanks (Process)	Nos.	6	6	6	6	6	6	6	6	6
Screen Chamber Dia x H	mm	600x1200	750x1500	900x1800	1000x2000	1200x2000	1400x2000	1400x2200	1600x2500	1800x2500
OGT Dia x H	mm	800x1500	900x1800	1000x2000	1200x2000	1400x2200	1600x2200	1600x2500	1800x2500	2000x2500
Equalization Tank Dia x H	mm	800x1500	1000x1800	1200x2000	1400x2200	1600x2200	1600x2500	1800x2500	2000x3000	2000x3000
Anoxic Tank Dia x H	mm	800x1500	1000x1800	1200x2000	1400x2200	1600x2200	1600x2500	1800x2500	2000x3000	2000x3000
Aeration Tank (MBBR) Dia x H	mm	1000x1800	1200x2000	1600x2000	1800x2500	2000x2500	2200x2500	2200x3000	2400x3000	2600x3000
Secondary Clarifier Dia x H	mm	800x1500	1000x1800	1200x2000	1400x2200	1600x2200	1800x2500	1800x2500	2000x3000	2200x3000
PSF Dia x H	mm	600x1500	750x1800	900x1800	1000x2000	1200x2000	1200x2200	1400x2200	1600x2500	1800x2500
ACF Dia x H	mm	600x1500	750x1800	900x1800	1000x2000	1200x2000	1200x2200	1400x2200	1600x2500	1800x2500
Treated Water Tank (TTP)	KL	3	5	10	15	20	25	30	50	60
Connected Load (Approx.)	kW	1.5	2.5	4.0	5.5	7.0	8.5	10.0	15.0	18.0
Total Area Required (L x W)	m	3.0 x 1.5	4.0 x 1.8	5.5 x 2.0	6.5 x 2.0	7.5 x 2.5	8.5 x 2.5	9.0 x 2.5	12.0 x 3.0	14.0 x 3.0

Note: Values are indicative and may vary based on inlet quality, applicable discharge norms and site conditions.

TYPICAL BILL OF MATERIAL (BOM) – 10 KLD (FRP PACKAGE STP)

Sr. No.	Equipment	Specification (FRP)	Qty.
1	Screen Chamber	900 x 1800	1
2	OGT	1000 x 2000	1
3	Equalization Tank	1200 x 2000	1
4	Anoxic Tank	1200 x 2000	1
5	Aeration Tank (MBBR)	1600 x 2000	1
6	Secondary Clarifier	1200 x 2000	1
7	MBBR Media	PE (Protected Surface Area)	1 set
8	Air Blower (1 + 1)	Root Blower	1 set
9	PSF	900 x 1800 (FRP)	1
10	ACF	900 x 1800 (FRP)	1
11	Treated Water Tank (TTP)	10 KL (FRP)	1
12	Sludge Transfer Pump	Submersible (1 + 1)	1 set
13	Filtration Feed Pump	Centrifugal (1 + 1)	1 set
14	Dosing System (Disinfection)	UV / Ozonation	1 set
15	Piping, Valves & Fittings	UPVC / CPVC	1 set
16	Electrical & Control Panel	MCC with PLC (Optional)	1 set

MOC (MATERIAL OF CONSTRUCTION)

Item	Material
Process Tanks (Standard)	FRP (Aquen)
Piping (Process)	UPVC / CPVC (as per service)
Pumps	CI / SS-304 (as per duty)
Blowers	Root Blower / Turbo Blower
Diffusers	EPDM Fine Bubble Diffuser
MBBR Media	HDPE (Virgin)
Screens	SS-304
PSF & ACF Vessels	FRP (Aquen)
Disinfection System	UV / Ozonation (Optional)
Treated Water Tank (TTP)	FRP (Aquen)
Instrumentation	pH, DO, Flow Meter, Level Switch, MLTSS, etc.
Control Panel	MCC with PLC (Optional)

KEY FEATURES

- No Civil Construction (Complete FRP Package)
- Compact & Modular Design
- Quick Installation
- Low Operation & Maintenance
- Odour Free Operation
- Reliable & Stable Performance
- Suitable for Residential, Commercial, Institutional & Industrial Applications
- Treated Water Tank for Flushing Purpose (TTP)
- Can be Installed Open to Sky or Underground
- Customized Design & Capacity
- Optional UV / Ozonation
- Online Monitoring (Optional)

INDUSTRIES & SECTORS WE SERVE



WHY AQUEN ?

- Customized Design
- High Treatment Efficiency
- Quality FRP Tanks & Components
- Cost Effective
- Competitive Pricing
- After Sales Support & AMC

AQUEN FLUID TECHNOLOGIES LLP

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WTP | STP | ETP | ZLD | FABRICATION | SPARES | AMC | RETROFIT | AUTOMATION

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